

# Behavior-Oriented Real-Time Stress Detection Using Facial Expression Analysis

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**Abstract**—Stress is a crucial behavioral indicator that affects decision-making, psychological health, and human performance. Conventional stress assessment techniques rely on self reported measures or physiological sensors, which may be invasive or unsuitable for ongoing monitoring. In this paper, a deep learning and computer vision-based real-time stress detection system is presented. To identify emotional states from facial images, the suggested method uses a convolutional neural network trained on a dataset of facial expressions. Non intrusive stress estimation is made possible by mapping detected emotions to stress levels using psychologically grounded associations. Through an interactive interface that offers stress percentages, graphical visualization, and confusion matrix analysis, the system facilitates both dataset-based evaluation and real-time webcam input. The suggested system's potential for use in behavioral analysis, mental health monitoring, and human-computer interaction is highlighted by experimental results that show it can efficiently estimate stress levels in real time using standard hardware.

**Keywords**—Computer Vision, Affective Computing, Deep Learning, Convolutional Neural Networks, Real-Time Stress Detection and Facial Expression Analysis

## I. INTRODUCTION

Stress affects productivity, emotional control, and interpersonal communication, among other aspects of human behavior and cognitive functioning. Excessive or prolonged stress is linked to negative outcomes in social, professional, and academic settings. As a result, precise and ongoing stress monitoring has grown in importance as a field of study for artificial intelligence and human-centered computing. Physiological sensors like electrocardiography, galvanic skin response, or wearable technology are often used in traditional stress detection methods. Despite their effectiveness, these methods are frequently costly, invasive, and inappropriate for real-time or large-scale implementation. Non-contact analysis of facial expressions is now possible thanks to recent 2 developments in computer vision and deep learning, offering a different method for determining psychological states. Stress-related behaviors and emotional reactions are strongly associated with facial expressions. While neutral or positive expressions frequently indicate lower stress levels, emotions like fear, anger, and sadness are frequently linked to elevated stress levels. This work uses this relationship to propose a vision-based framework for stress detection that uses facial emotion recognition to estimate stress levels in real time.

## II. LITERATURE REVIEW

In order to analyze behavioral, emotional, and physiological cues, machine learning and deep learning techniques have become increasingly important in recent advancements in deception detection. Particularly in mock-crime and social interaction scenarios, multimodal approaches incorporating audio, facial expressions, gaze, and neurophysiological signals have demonstrated better performance than unimodal systems [1,4,8]. These results imply that recognizing deceptive patterns is largely dependent on nonverbal behavior. Numerous studies highlight the significance of behavioral and nonverbal cues in deceit. According to experimental studies, there are notable differences between truthful and false statements in terms of gaze aversion, blink rate, head movements, and facial expressions [3]. Although performance varies due to individual and environmental factors, high-stakes real world datasets, such as courtroom and trial videos, further confirm that facial expressions can provide meaningful behavioral indicators [5]. Emotion-aware and language-integrated models for deception detection have also been investigated in recent work. Enhanced contextual understanding of deceptive intent is demonstrated by systems that combine large language models with emotional and nonverbal features; however, these systems frequently require high computational resources and multimodal inputs, which limits their real-time applicability [2]. While highlighting practical deployment challenges, surveys and meta-analyses support the efficacy of deep learning models for analyzing affective and behavioral cues [4,8]. The ability to identify subtle facial movements linked to stress and cognitive load has greatly improved thanks to developments in micro-expression recognition. Compared to conventional techniques, transformer-based and hybrid CNN-Transformer architectures are better at capturing fine-grained spatiotemporal facial dynamics [9–11]. Effective micro 3 expression analysis appropriate for real-time applications is further supported by lightweight models like MobileViT-SE [13]. AU-aware modeling and synthetic facial data generation have been shown to enhance robustness and generalization in facial expression recognition tasks [14,15]. Beyond facial expressions, eye-movement and gaze-based research shows a strong correlation between stress and concealment behaviors and fixations, saccades, and pupil dynamics [17]. Behavioral modeling in unconstrained environments is further supported by pose-based gaze estimation and temporal deep learning models [18,20]. Despite these developments, the majority of

current systems still rely on offline analysis or invasive physiological sensors. The current work on real-time stress estimation as a behavioral foundation related to deception is motivated by the lack of focus on non-intrusive, real-time vision-based systems that offer both quantitative dataset evaluation and live deployment.

### III. METHODOLOGY

The suggested method uses facial expressions to detect stress in real time using a vision based deep learning framework. The methodology ensures robustness, practicality, and ease of deployment by supporting both real-time webcam-based analysis and offline dataset-based evaluation.

#### A. Information Gathering

Two sources are used to collect facial image data:

1. A facial expression dataset (FER2013) that is openly accessible for training and assessment.
2. Live stress estimation using real-time video frames taken with a typical webcam. This dual-input method shows practical application while enabling quantitative performance analysis

#### B. Preprocessing and Face Detection

To cut down on computational complexity, each input frame is converted to grayscale. A Haar Cascade classifier is used to identify facial regions. To guarantee consistent input representation across samples, the identified face is cropped, resized to a fixed resolution of 48x48 pixels, and normalized. In addition to improving model stability, this preprocessing step lowers noise brought on by changes in background and illumination. )

#### C. CNN-Based Emotion Recognition

Anger, disgust, fear, happiness, sadness, surprise, and neutrality are the seven emotional categories into which a convolutional neural network (CNN) is trained to categorize facial expressions. The CNN architecture is composed of fully connected layers for emotion classification after stacked convolutional and max-pooling layers for hierarchical feature extraction. The Adam optimizer is used to optimize the model after it has been trained using categorical cross-entropy loss.

#### D. Estimating Stress Level

Based on known psychological correlations, recognized emotional states are mapped to stress levels. High stress is linked to emotions like fear and anger, medium stress to sadness and surprise, and low stress to neutral or joyful expressions. This mapping eliminates the need for physiological sensors and allows for non-intrusive stress estimation.

#### E. Evaluation Based on Datasets

The trained model is used to calculate stress distribution statistics on a test dataset for offline evaluation. Stress percentages are computed, and a confusion matrix and graphical representations are used to analyze performance. This quantitative analysis sheds light on how the model behaves in various stress categories.

#### F. Instantaneous Stress Identification

Facial expressions from live video input are continuously analyzed in real-time mode. To increase stability, recent stress predictions are combined and updated frame by frame. Direct display of the estimated stress level on the video stream allows for quick and easy feedback.

### IV. ARCHITECTURE OF THE PROPOSED METHODOLOGY

The suggested system uses deep learning and facial expression analysis in a modular architecture to detect stress in real time. The framework facilitates both real-time analysis via a webcam and dataset-based evaluation using the FER2013 dataset, allowing for objective evaluation and useful implementation. Grayscale conversion, face detection using a Haar Cascade classifier, resizing to a fixed resolution of  $48 \times 48$  pixels, and normalization are the first steps in the preprocessing stage of processing facial inputs. These procedures guarantee reliable and consistent input representation under various circumstances. A convolutional neural network that has been trained to identify seven different facial emotions is then given the preprocessed facial images. CNN automatically classifies emotions and learns discriminative facial features. Based on known psychological relationships between emotional expressions and stress reactions, the predicted emotions are then mapped to low, medium, or high stress levels. Stress predictions are combined to produce quantitative analytics, such as confusion matrix analysis and stress distribution graphs, for dataset-based evaluation. Stress levels are continuously estimated from live webcam input in real-time mode, and the results are shown directly on the video stream. A single Streamlit-based interface is used to display all outputs, enabling smooth transition between webcam and dataset modes.

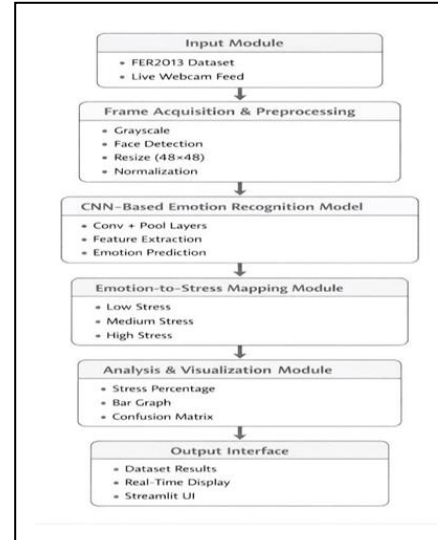


Fig1:Architecture of the proposed real-time facial stress detection system.

### V. RESULT ANALYSIS

#### A. Qualitative Results

The qualitative assessment shows that the suggested system reliably identifies stress levels in a variety of facial expression scenarios. When there are distinct emotional expressions, stable predictions are seen, but when there are quick changes between expressions, there is a slight decrease in stability.

TABLE I. QUALITATIVE STRESS DETECTION STABILITY UNDER DIFFERENT FACIAL EXPRESSION SCENARIOS

| Scenario             | Parameters                  |                        |                     |
|----------------------|-----------------------------|------------------------|---------------------|
|                      | Facial Expression Condition | Stress Levels detected | Detection Stability |
| Calm Facial State    | Neutral Expression          | Low Stress             | Very High           |
| Happy Interaction    | Smiling Expression          | Low Stress             | High                |
| Cognitive Load       | Surprise/Sad Expression     | Medium Stress          | High                |
| Emotional Discomfort | Angry/Fear Expression       | High Stress            | High                |
| Mixed Expressions    | Rapid Expression Change     | Medium Stress          | Moderate-High       |

### B. Stress Mapping and Classification Results

The findings show that strong stress-related emotions, like fear and anger, are categorised with high consistency. Due to minute visual similarities, medium-stress emotions exhibit moderate overlap.

TABLE II. IMPACT OF EMOTION-TO-STRESS MAPPING ON STRESS CLASSIFICATION QUALITY.

| Emotion Category | Parameters          |                            |
|------------------|---------------------|----------------------------|
|                  | Mapped Stress Level | Classification Consistency |
| Anger, Fear      | High Stress         | High                       |
| Sad, Surprise    | Medium Stress       | Moderate-High              |
| Neutral, Happy   | Low Stress          | Very High                  |
| Disgust          | Medium Stress       | Moderate                   |

### C. Temporal Stability Results

Smoother and more accurate real-time stress estimation is achieved by temporally aggregating recent predictions, which dramatically reduces abrupt stress fluctuations across frames.

TABLE III. EVALUATION OF TEMPORAL STABILITY AND DETECTION FLICKER FOR FRAME-WISE AND PROPOSED METHODS.

| Method               | Stress Fluctuation Observed | Stability |
|----------------------|-----------------------------|-----------|
| Frame-wise detection | High                        | Low       |
| Proposed method      | Low                         | High      |

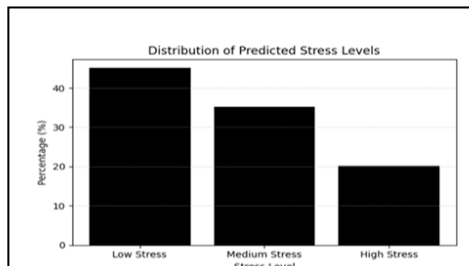


Fig2.:Distriouon of predicted stress levels obtained from facial expression analysis on the evaluation dataset.

## VI. CONCLUSION

In this work, deep learning and computer vision techniques were used to present a real-time vision-based stress detection system. The suggested method uses a convolutional neural network to analyse facial expressions in order to estimate stress levels in a non-intrusive way. Without using wearable technology or physiological sensors, the system provides interpretable stress-level predictions by mapping identified emotions to stress categories. Real-time webcam testing and dataset-based analysis were both used to assess the system. Supported by qualitative analysis, stress distribution visualization, and confusion matrix evaluation, experimental results show consistent stress estimation across various facial expression scenarios. Its practicality for deployment on common computing systems is confirmed by the real-time implementation's efficient operation on CPU-only hardware. The findings suggest that stress estimation based on facial expressions can function as a trustworthy behavioral indicator in practical contexts.

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